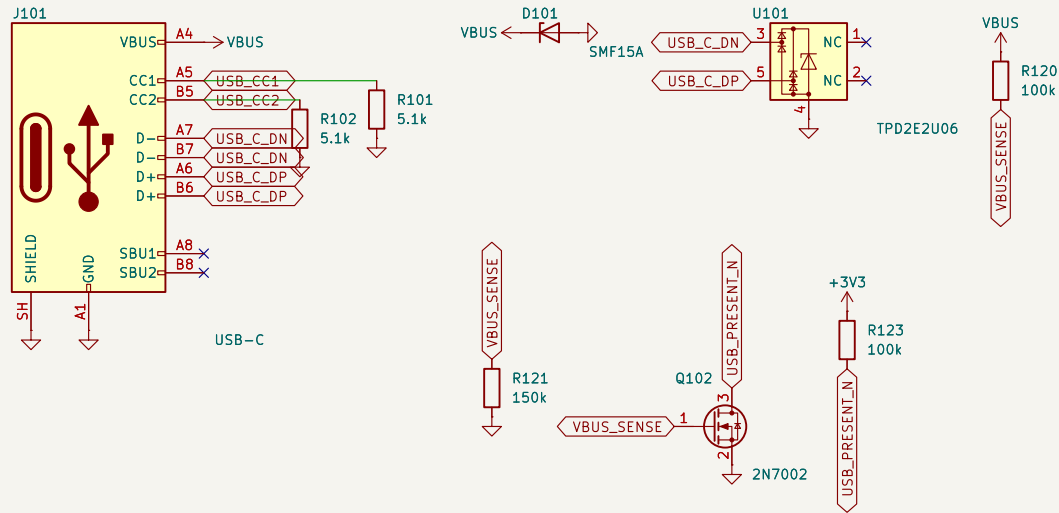




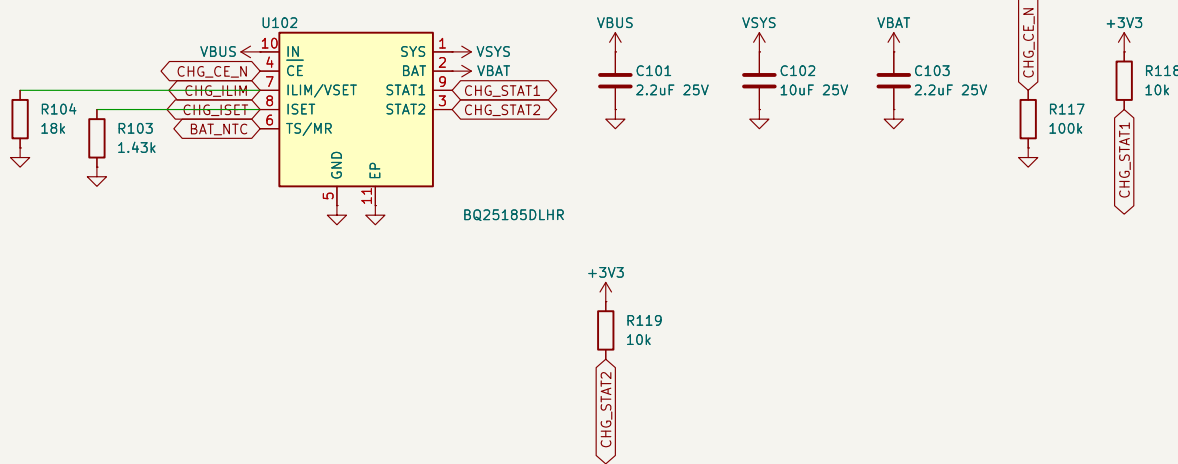
POWER

USB-C ENTRY



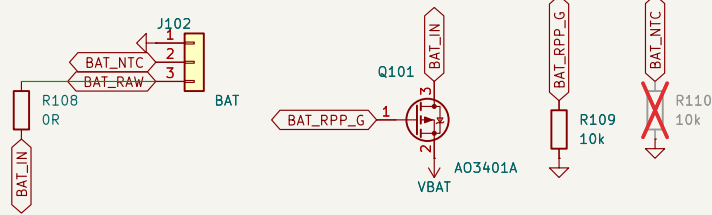
Sink only; 5.1k Rd on CC1/CC2 (ODD JOBS 25); Default USB power, 500 mA. ESD at the connector (119); TPD2E2U06 on D+/D-, SMF15A on VBUS. Shell to GND (plastic enclosure, 26). The 22R series resistors sit at the module (compute sheet). VBUS_SENSE 100k/150k (5.0 V -> 3.0 V, 0 uA without USB) drives the gate of Q102; its drain is USB_PRESENT_N, active low, with a 100k pull-up to +3V3 (R123, 33 uA only while USB is present), on the RTC pad GPIO2, so plugging USB wakes A1 from deep sleep (ext1 any-low; design review 2026-10-06). 2N7002: VGS(th) 2.5 V max at 250 uA against >= 2.85 V of gate at the USB minimum 4.75 V; VGS 20 V max, so a faulty source up to 33 V leaves the gate intact.

CHARGER



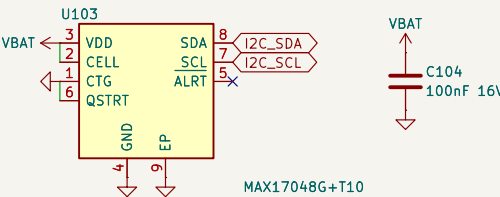
BQ25185 linear charger with power path: runs MAO from USB while charging. SYS 4.5 V on USB. ILIM/VSET R104 18k = 4.2 V, 500 mA input limit (SLUSF65B table 6-1). ISET R103 1.43k 1% = 300 AOhm / 1.43k = 210 mA charge (199-220 mA), as AO (the LP503035 cell allows 250 mA, 0.5C). TS/MR to the cell's 10k NTC (DNP 10k fallback R110); the fixture enters factory (ship) mode by holding TS/MR low 10 s with USB present (TP TSMR). /CE from GPIO18 (CHG_CE_N, 100k pull-down; charging on from reset); firmware pauses charging outside the cell's 0-45 C. STAT1/STAT2 open-drain, 10k pull-ups to +3V3, to GPIO33/34; high-Z on battery, 0 uA. BATFET off below 3.0 V (BUVLO); the hardware floor that replaces AO's UVLO divider. Caps per TI: IN and BAT 2.2 uF 25 V 0402 (>= 1 uF left after DC bias, SLUSF65B 7.2.2.3), SYS 10 uF 25 V.

BATTERY



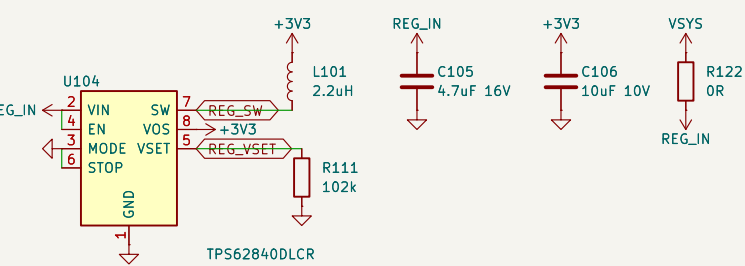
JST SH 3-pin: 1 = BAT-, 2 = NTC, 3 = BAT+ (same order as A0 and KINO J1100, ODD JOBS 41/167). Kept from A0: the Gate C JST PH (2 A) is 4.8 mm tall and does not fit the 3.2 mm zone over the cell (owner decision, A1 report). Cell MUST carry its own protection PCM and a 10k NTC. R108 = LINK_BAT, OR to measure battery current (110). Q101 blocks a reversed cell (51); its drop is in the low-battery budget (A1 report). R110 fits only for cells without NTC.

FUEL GAUGE



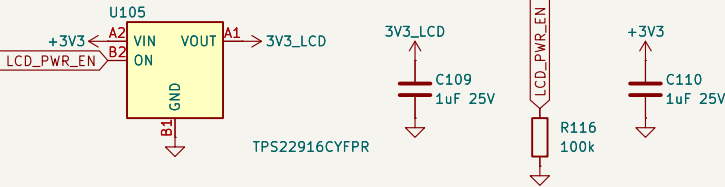
MAX17048 ModelGauge on VBAT, I2C 0x36, ALERT not connected: firmware polls SOC (Gate C).

CORE 3V3 BUCK



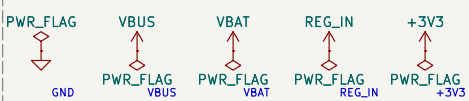
TPS62840 from REG_IN (= VSYS through LINK_REG R122): 750 mA, 60 nA Iq. PFM (MODE low), STOP low, EN to VIN. VSET R111 102k 1% = 3.2 V (SLVSEC60 table 1, TPS62840: 97.92-106.08k); 0.1 V under the panel VCI 3.3 V maximum. Below -3.25 V in it runs in 100% mode (output = Input - I x (RDS_on + DCR)). L 2.2 uH DFE201612E-2R2M, CIN 4.7 uF, COUT 10 uF; with the loads the +3V3 rail stays under 40 uF effective. No UVLO divider: the charger's 3.0 V BATFET cut-off and a firmware floor replace it.

DISPLAY RAIL



TPS22916C switches 3V3_LCD (panel logic and the backlight op-amp; the LEDs run from VSYS). ON from GPIO6 with its 750k smart pull-down and a 100k: off until firmware enables it. Slow rise (0.9 ms at 3.6 V), 150 ohm quick output discharge: the rail really reaches 0 V when off, 10 nA off.

RAIL SOURCES



USB-C entry, charger with power path, battery, fuel gauge, 3.2 V buck, display rail
ODD JOBS

Sheet: /Power/
File: power.kicad_sch

Title: MAO_MAIN A1 - Power

Size: A3
KiCad E.D.A. 10.0.6

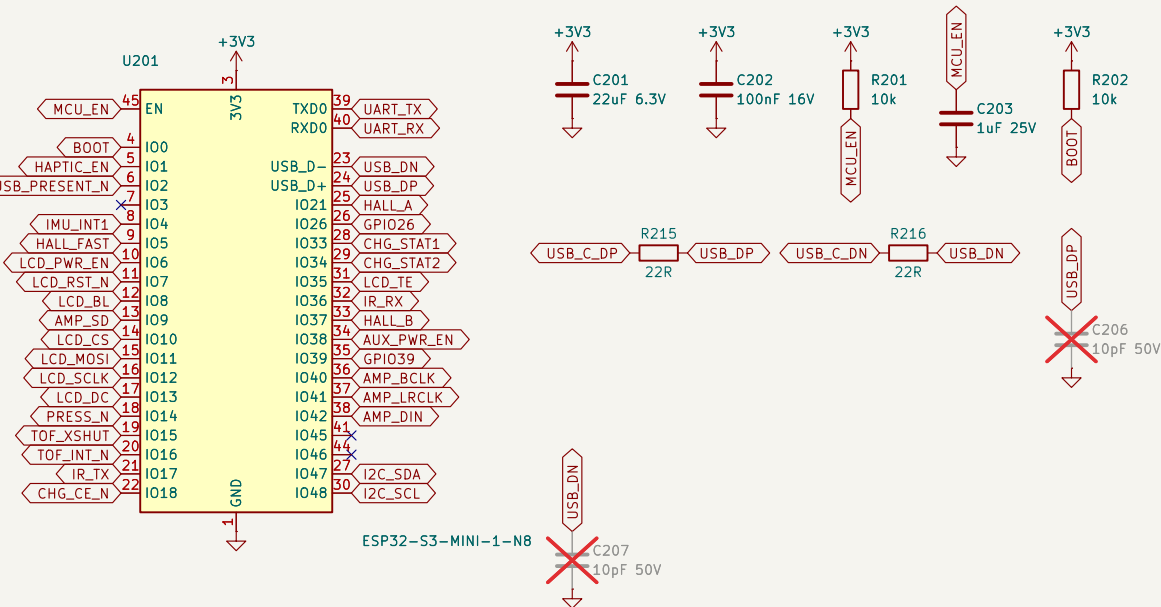
Date: 2026-10-06

Rev: A1

Id: 2/6

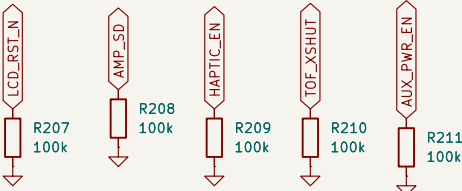
COMPUTE

MCU



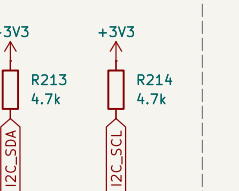
ESP32-S3-MINI-1-N8 (8 MB quad flash, no PSRAM, 85 C) on B.Cu at 6 o'clock, antenna over the board-edge notch (no copper beneath, Espressif keep-out), 22 uF + 100 nF at 3V3 pad 3. EN: 10k/1 uF RC. GPIO0 = BOOT pad only. GPIO3/45/46 NC. Spares GPIO26/39 on test pads, GPIO43/44 on the Tag-Connect.

DEFAULTS



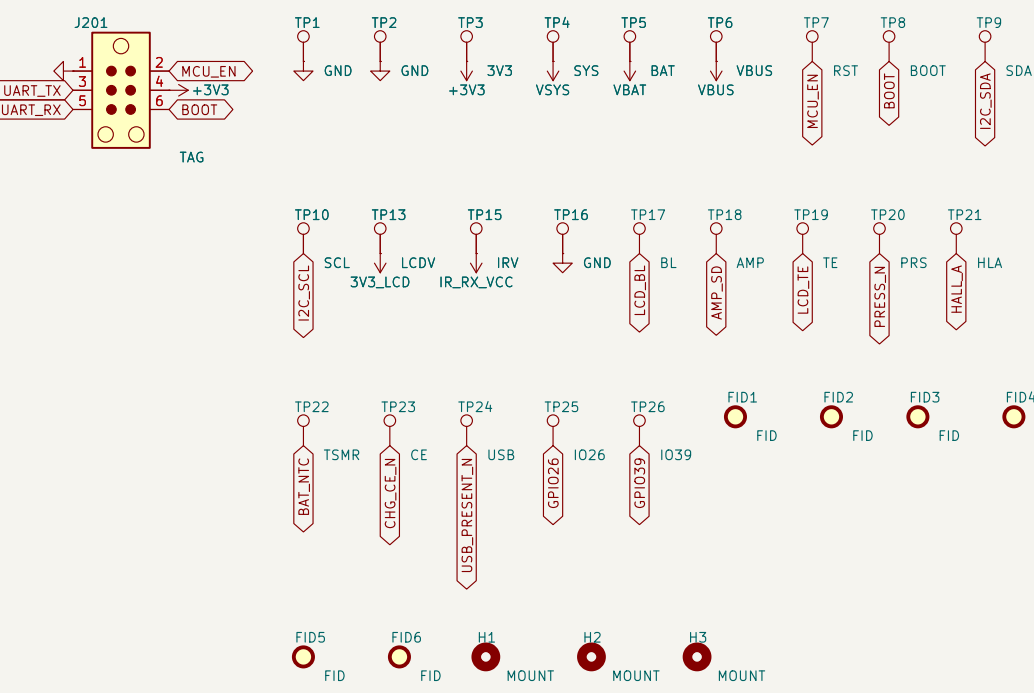
Hardware pull-downs that hold every peripheral off from reset and in deep sleep (ODD JOBS 116, 117). Each sits at the pin it serves.

I2C



One 400 kHz bus, four devices (IMU, ToF, gauge, haptic), all on always-on rails. 4.7k pull-ups to +3V3 (Gate C), one pair only.

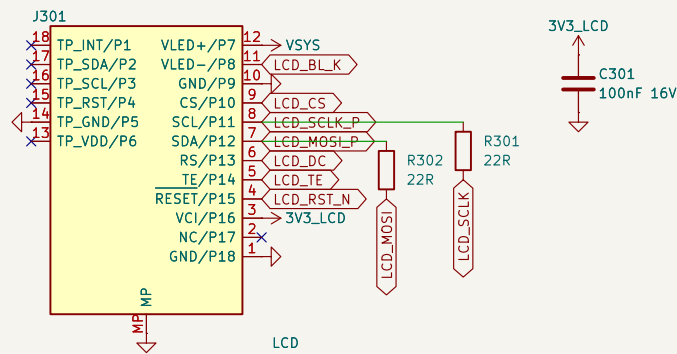
SERVICE



Tag-Connect TC2030-NL (no part fitted): 1 GND, 2 EN, 3 TXD0, 4 3V3, 5 RXD0, 6 GPIO0 BOOT - recovery without USB (ODD JOBS 33-35, 157). Probe pads: rails beside the power section, comms and wake lines in one field, switched rails at their sources (36-39).

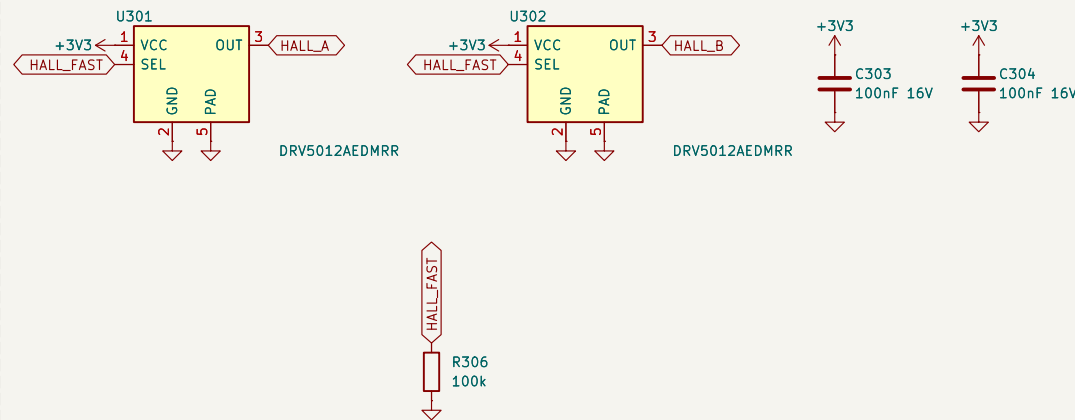
INTERFACE

DISPLAY



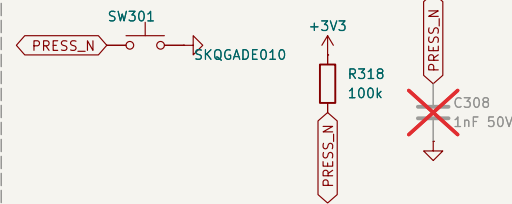
Plug-in display: 1.28" round GC9A01 panel (Winstar WF0128B7YA4AANDN0, 240x240 IPS) on an 18-pin 0.5 mm FPC tail, folded under the panel into a 1.0 mm high back-flip connector with top and bottom contacts, so pins 1–6 of this 18-pin standard plugs in either way and can be swapped without solder. Pins 4–6 serve the touch controller of touch variants: not wired, TP_GND to GND. TE to GPIO35 for tear-free frames. 22R on SCLK/MOSI demands the edges (ODD J0B5 29). Panel logic on the switched LV3_LCD rail, LEDs from VSY5 through the current sink; RST held low by 100k until firmware. Every line to the panel is low or Hi-Z while V3_LCD is off (firmware, Gate C). The stock 70 mm tail lies as one loop under the panel, drops through the board slot at 9 o'clock and plugs into J301 on B.Cu.

RING DIAL



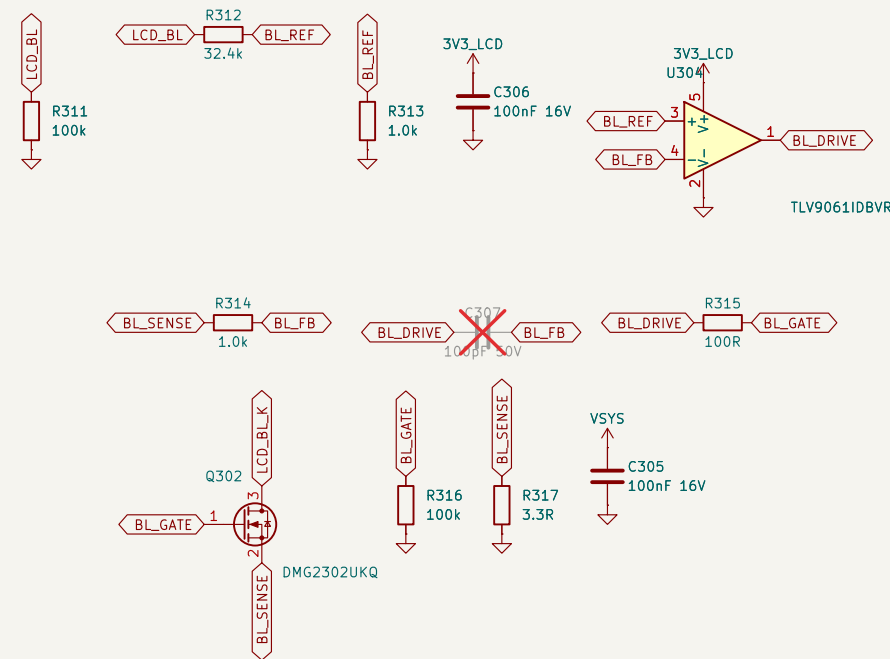
Two DRV5012 Hall latches under the ring's 30-pole ferrite strip, 6 deg apart (half a pole = 90 deg electrical): A/B quadrature identical to the LCDkit EC11 (2 transitions per detent, 30 detents). HALL_A on GPIO21 (RTC: deep-sleep wake, ext0 armed at the opposite level), HALL_B on GPIO37. HALL_FAST (GPIO5) low = 20 Hz sampling, 1.3 uA (sleep); high = 2.5 kHz (awake); 100k pull-down.

FACE PRESS



ALPS SKQG on F.Cu at the centre, under the panel: the floating face presses it. PRESS_N on GPIO14 (RTC: deep-sleep wake, ext1 any-low), not a strap, 100k pull-up draws only while pressed; DNP 1 nF makes an optional 100 us RC.

BACKLIGHT



Constant-current low-side sink on VVSYS (Gate C): LCD_BLD (GPIO8, LEDC -30 kHz) $\rightarrow$ 32.4k / 1.0k $\rightarrow$ BL_REF (0 / 95.8 mV at 3.2 V; 3.2 V $\times$ 100 / 33.4k); TLV9061 on 3V3_LCD (0 uA in deep sleep) drives DVG3302UQO so that BL_SENSE follows BL_REF: Rs 3.3 ohm 1k = 29.0 mV at 100% (Gate C1 at 3V3_LCD at 3.3 V). Regulates for VVSYS $\rightarrow$ DMF (3.4 V max) + 0.1 V. Off from reset: LCD_BLD 100k, BL_REF 1.0k, Gate 100k to GND, op-amp unpowered with the panel, DNP 100 pF load compensation (fit if the sink rings). Rs is re-set when Winstar's backlight data arrives.

Round display, backlight current sink, ring dial, face press

ODD JOBS

Sheet: /Interface/

File: interface.kicad_sch

Title: MAO_MAIN A1 – Interface

Size: A3

Date: 2026-10-06

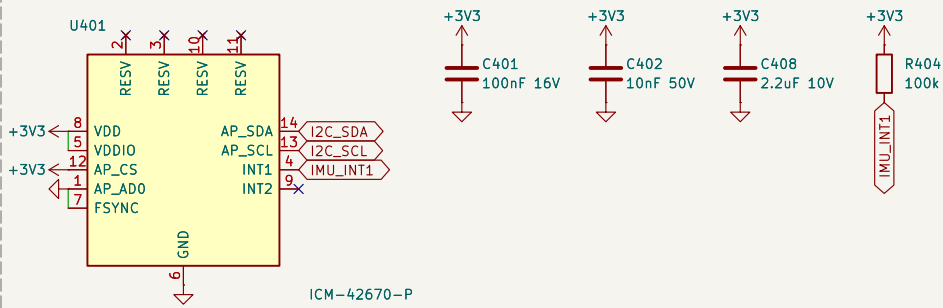
Rev: A1

STEEP AS	
KiCad E.D.A. 10.0.6	

Id: 4/6

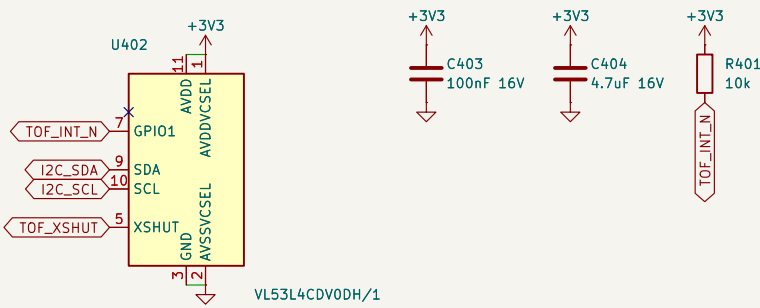
SENSE

IMU



ICM-42670-P at 0x68 (AP_AD0 low), I2C (AP_CS to VDDIO), FSYNC to GND, RESV pins open (DS-000451 table 9), INT1 (GPIO4, RTC) = wake-on-motion: open-drain active-low, 100k pull-up. INT2 not wired (pin map). VDD 0.1 uF + 2.2 uF X7R, VDDIO 10 nF X7R (table 10). On F.Cu: +Z out of the face.

PROXIMITY



VL53L4CD time-of-flight, 0-1.3 m, under the window border. XSHUT from GPIO15, 100k pull-down (off by default); GPIO1 open-drain to GPIO16 with the 10k pull-up ST recommends. AVDD decoupled 4.7 uF + 100 nF per ST.

IMU, proximity

ODD JOBS

Sheet: /Sense/

File: sense.kicad_sch

Title: MAO_MAIN A1 - Sense

Size: A3

Date: 2026-10-06

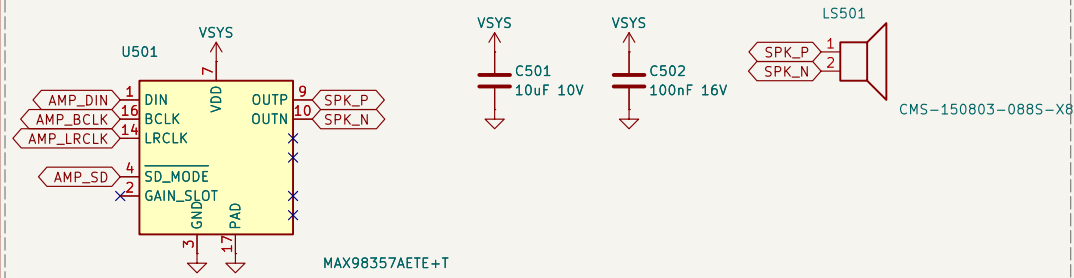
Rev: A1

KiCad E.D.A. 10.0.6

Id: 5/6

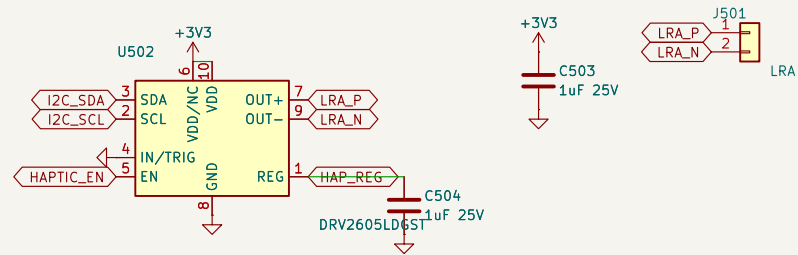
FEEDBACK

SPEAKER AMP



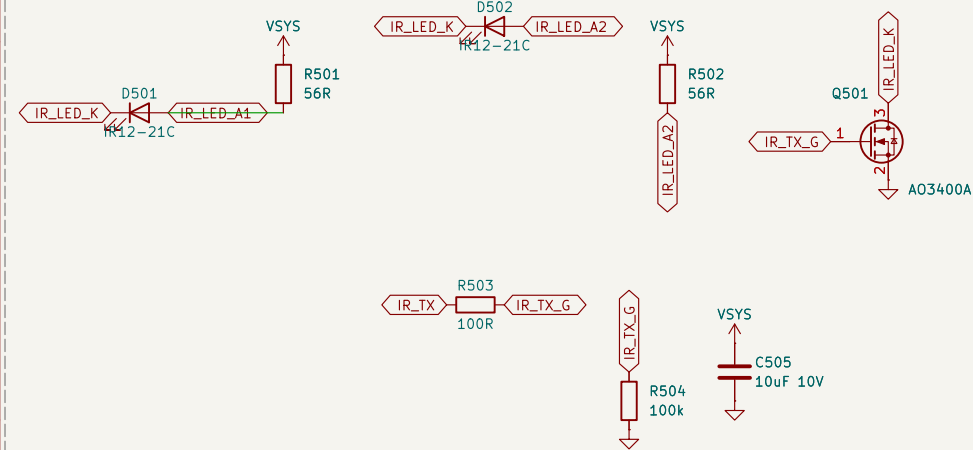
MAX93575A I2S class-D from VSYS (Gate C). SD_MODE driven directly from GPIO9 (AMP_SD): high = left channel, low = 0.6 uA shutdown; 100k pull-down plus the internal 100k; silent from reset (ODD JOBS 116). AO's 2.2k series resistor is gone: its datasheet reason (logic high above VDD) cannot occur, +3V3 < VSYS by topology. I2S on GPIO40/41/42. GAIN_SLOT open = 9 dB. Speaker: Same Sky CMS-150803-0085-XB (15 x 3 x 3 mm, 8 ohm, 0.8 W) in the base at 9 o'clock; its own spring contacts press on two pads on B.Cu (LS501).

HAPTIC



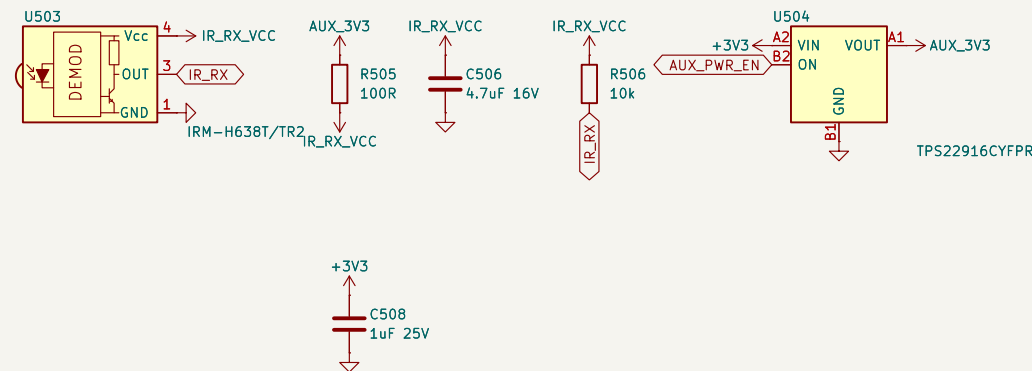
DRV2605L closed-loop LRA driver on +3V3 (Gate C: its SDA/SCL/EN/TRIG absolute maximum is VDD + 0.3 V, so it shares the always-on-rail of the I2C pull-ups and is never supply-gated). VDD pin 10 and VDD/NC pin 6 to +3V3. I2C 0x5A, EN from GPIO1 with 100k pull-down (4 uA off). IN/TRIG low: playback by I2C. REG 1 uF, VDD 1 uF. LRA LD0832AA (235 Hz, 1.8 Vrms) glued to the base, leads to J501.

IR TRANSMIT



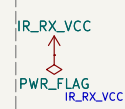
Two side-emitting 940 nm LEDs at the back edge (B,Cu), each with its own 56R from VSY5 (26–59 mA pulses at 38 kHz, 33 % duty: under the 65 mA rating at any VSY5 up to 4.5 V and any Vf). A03400A low side driven from GPIO17 through 100R; the 100k at the gate is also IR_TX's pull-down: dark at reset (117).

IR RECEIVE



IRM-H638T (as LCDkit), top-view, looking up through the window border, on GPIO36. Powered from AUX_3V3, a TPS22916C on +3V3 switched by AUX_PWR_EN (GPIO38), through 100R/4.7 uF only while listening (0.4 mA). OUT pull-up goes to the switched supply, never to +3V3, so an unpowered receiver is never back-powered and reads a defined low while it is off.

RAIL SOURCES



Speaker amplifier, haptic driver, IR transmit and receive

ODD JOBS

Sheet: /Feedback/
File: feedback.kicad_sch

Title: MAO_MAIN A1 – Feedback

Size: A3

Date: 2026-10-06

Rev: A1

Size: A5	
KiCad E.D.A. 10.0.6	

Id: 6/6